# BIM493 Mobile Programming I Assignment #2

## Smart Home Controller App

In this assignment, you are asked to build a simple **Smart Home Controller App** in Flutter. The app should allow the user to view and interact with different types of smart home devices. Each device should display basic information and a device-specific status or action (for example, turning a light on/off, adjusting a thermostat, or playing/pausing a speaker).

Each smart home device must be modeled as a class by applying the object-oriented programming principles and Dart language features listed below.

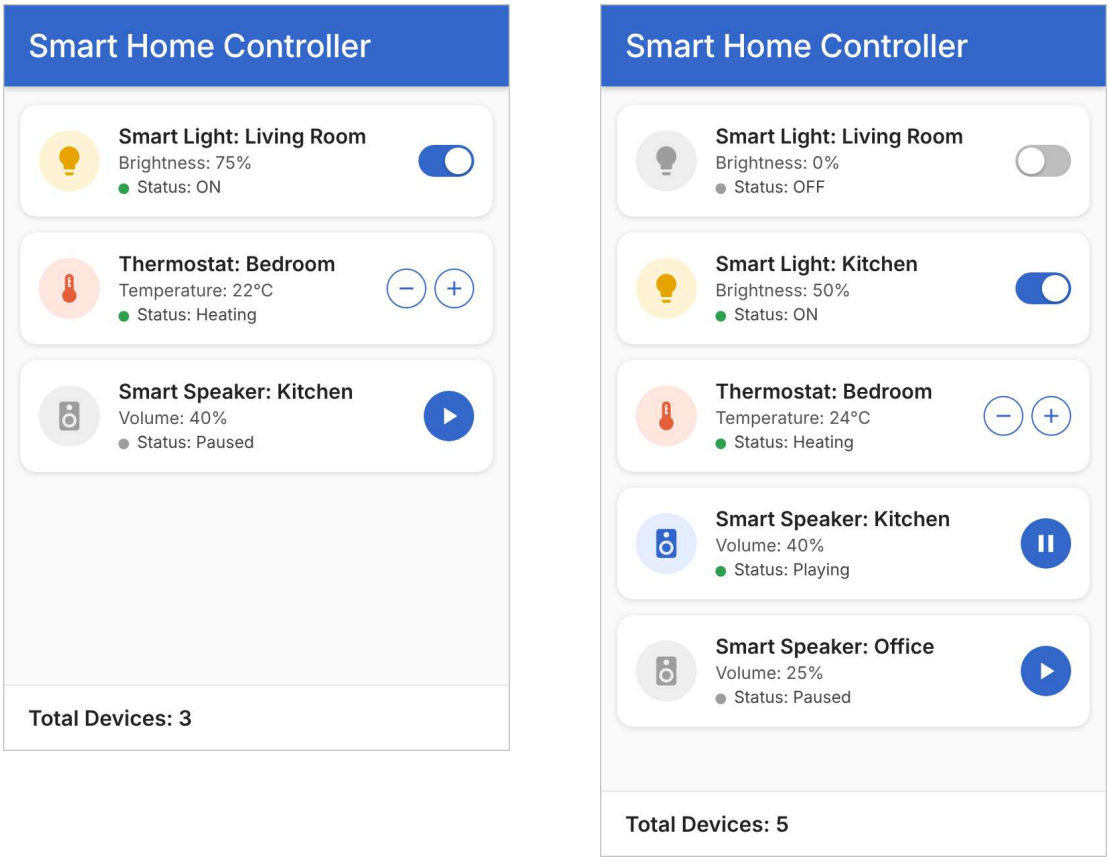
| Concept | Requirement |
|---|---|
| **Class** | Create a base class named **SmartDevice** and the subclasses **SmartLight**, **Thermostat**, and **SmartSpeaker**. |
| **Objects** | Create multiple device instances and display at least one instance of each subclass in the application. |
| **Constructor** | Use both default and named constructors when creating device objects. |
| **Inheritance** | Make **SmartLight**, **Thermostat**, and **SmartSpeaker** extend the base class **SmartDevice**. |
| **Abstraction** | Make **SmartDevice** an abstract class and define at least one abstract method, such as **getStatus()** or **performAction()**. |
| **Interface** | Define an interface (for example, **Connectable**) and implement it in the device classes. The interface should include at least one method. |
| **Mixin** | Create a mixin (for example, **Schedulable**) and use it in at least one suitable device class. |
| **Keywords** | Demonstrate the use of **static**, **super**, and **this** in your implementation. |

## Expected Output (UI Screen)

Your application interface should include the following elements:

- **Device List:** Display at least three device cards: Smart Light, Thermostat, and Smart Speaker.
- **Device Information:** Each card should show the device name, a relevant property (such as brightness, temperature, or volume), and its current status.
- **Interaction:** Provide at least one meaningful control or action for each device type (for example, a switch, a temperature adjustment control, or a play/pause button).
- **Bottom Text:** Display the total number of created devices. This value must come from a **static variable** in your class implementation.

You may choose your own colors, icons, and layout. However, all required classes and Dart language features must be implemented in the application; the interface alone is not sufficient.




*Sample screens – colors, icons and layout may differ.*

## Submitting the Assignment

- You would create a GitHub account and submit your files to be publicly available to the community **or** minimize your Project and upload to a file hosting service like Google Drive, Microsoft OneDrive or Dropbox etc. if you do not want to share your code with others.
- Your submission must contain the Flutter project source code. Check that the project can be opened and run before submitting it.
- Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.

## Due Date

**November 10, 2026, Tuesday – 23:59**

Late submission is not allowed. Submissions via e-mail will not be accepted.